

Given Problem

তোমার ফাংশনটা হলো:

$$f(n) = \begin{cases} 0 & \text{if } n = 0 \\ 1 & \text{if } n = 1 \\ 2 & \text{if } n = 2 \\ 2f(n-1) + f(n-3) & \text{if } n \geq 3 \end{cases}$$

Final Exam Answers

◆ (i) Find $f(4)$

Given:

$$\begin{aligned} f(0) &= 0, \quad f(1) = 1, \quad f(2) = 2 \\ f(3) &= 2f(2) + f(0) = 2 \times 2 + 0 = 4 \\ f(4) &= 2f(3) + f(1) = 2 \times 4 + 1 = 9 \end{aligned}$$

 Answer:

$$f(4) = 9$$

◆ (ii) Time Complexity

Recurrence:

$$T(n) = T(n-1) + T(n-3)$$

Each call generates two recursive calls $\rightarrow$ recursion tree.

- Branching factor $b = 2$
- Height $h \approx n$

Total nodes:

$$2^n - 1 \approx 2^n$$

 Time Complexity:

$$O(2^n)$$

Optimal Substructure:

The value of $f(n)$ depends on smaller subproblems $f(n-1)$ and $f(n-3)$.
So, the problem can be broken into smaller subproblems.

Conclusion:

Since the problem has:

- Overlapping subproblems
- Optimal substructure

Therefore, it can be solved using **Dynamic Programming**.
